



Q-SOLVE KENYA 2026

Quantum Computing Hackathon Judging Tool

Judging and Assessment Instrument

1st–3rd September 2026

Strathmore University, Nairobi, Kenya

1.0 Criteria at a Glance

Code	Criterion	Maximum Score
C1	Problem Understanding and Local Relevance	15
C2	Quantum Computing Relevance and Necessity	20
C3	Technical Correctness and Scientific Integrity	15
C4	Evidence, Benchmarking and Validation	15
C5	Innovation and Intellectual Contribution	12
C6	Sustainable Development Impact (SDGs 2, 3 and 8)	13
C7	Feasibility, Scalability and Adoption	10

2.0 Detailed Judging Criteria

Criterion	Score	Performance Standard / Assessment Guidance
2.1 C1 – Problem Understanding and Local Relevance (Maximum 15 points) <i>Is the problem real, specific and grounded in local realities?</i>		
C1	13–15	Defines a specific, important problem grounded in Kenyan, East African or clearly identified local realities. Identifies users, current practice, constraints and consequences, supported by credible evidence or stakeholder insight.
C1	10–12	Presents a relevant, reasonably well-defined problem, but some assumptions or contextual details remain unverified.
C1	6–9	Addresses a relevant challenge, but the problem is broad, generic or insufficiently localised; user needs and operating conditions are only partly understood.
C1	1–5	Presents a weakly defined or technology-led problem with little understanding of users, institutions or local conditions.
C1	0	Does not address the selected challenge or track.
SCORE AWARDED: _____ / 15		
2.2 C2 – Quantum Computing Relevance and Necessity (Maximum 20 points) <i>Is quantum computing technically justified and necessary for this problem?</i>		

Criterion	Score	Performance Standard / Assessment Guidance
C2	17–20	Clearly identifies the computational bottleneck and explains why a quantum approach is appropriate. Maps the problem to a relevant quantum algorithm, circuit, annealing formulation, quantum simulation or workflow; includes a credible classical baseline and avoids exaggerated quantum-advantage claims.
C2	13–16	The quantum approach is technically plausible and reasonably explained, but comparative advantage, algorithmic mapping, resource analysis or baseline comparison is incomplete.
C2	8–12	Uses a recognisable quantum-computing technique, but the justification is superficial, generic or mainly conceptual.
C2	1–7	The quantum component appears unnecessary, decorative, technically misunderstood or weakly connected to the problem.
C2	0	No meaningful quantum-computing component is demonstrated.

SCORE AWARDED: _____ / 20

2.3 C3 – Technical Correctness and Scientific Integrity (Maximum 15 points)

Is the quantum methodology valid, transparent and reproducible?

C3	13–15	Methodology is technically coherent and scientifically defensible. Assumptions and limitations are explicit; results are reproducible or independently inspectable; circuit/formulation, data encoding and measurement choices are clearly explained.
C3	10–12	Implementation is mostly correct but contains minor technical gaps, undocumented assumptions or methodological weaknesses.
C3	6–9	Shows partial technical correctness, but validation, circuit details, parameters, implementation details or methodological explanations are incomplete.
C3	1–5	Contains serious conceptual errors, unverifiable results, misleading benchmarks or weak scientific reasoning.
C3	0	The approach is fundamentally invalid, deceptive or scientifically indefensible.

SCORE AWARDED: _____ / 15

2.4 C4 – Evidence, Benchmarking and Validation (Maximum 15 points)

Does the quantum solution demonstrably work?



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Criterion	Score	Performance Standard / Assessment Guidance
C4	13–15	Provides quantitative results, relevant baselines, appropriate metrics and a transparent evaluation procedure; tests uncertainty, robustness, sensitivity, noise or scalability where appropriate.
C4	10–12	Shows promising results, but benchmarking, validation, robustness testing, hardware comparison or sample size is limited.
C4	6–9	Presents a functioning prototype or demonstration, but evidence is narrow, anecdotal, simulation-only without context or based on a single example.
C4	1–5	Relies mainly on theoretical claims, mock outputs, unsupported projections or demonstrations that do not test the central hypothesis.
C4	0	Provides no meaningful evidence that the proposed quantum approach works.

SCORE AWARDED: _____ / 15

2.5 C5 – Innovation and Intellectual Contribution (Maximum 12 points)

Does the solution contribute something original or meaningful?

C5	10–12	Introduces a distinctive problem formulation, quantum method, encoding strategy, circuit design, system architecture, implementation model or application pathway, and explains how it differs from existing approaches.
C5	7–9	Provides a meaningful adaptation, integration or application of known quantum-computing methods in a new context.
C5	4–6	Demonstrates competent implementation but limited originality or intellectual contribution.
C5	1–3	Primarily repackages a tutorial, standard demonstration, existing repository or widely available solution.
C5	0	No identifiable original contribution.

SCORE AWARDED: _____ / 12

2.6 C6 – Sustainable Development Impact: SDGs 2, 3 and 8 (Maximum 13 points)

Does the solution provide a credible path from quantum technology to progress on SDG 2: Zero Hunger, SDG 3: Good Health and Well-Being, and/or SDG 8: Decent Work and Economic Growth?



Criterion	Score	Performance Standard / Assessment Guidance
C6	11–13	Clearly identifies one focus SDG; SDG 2: Zero Hunger; SDG 3: Good Health and Well-Being; or SDG 8: Decent Work and Economic Growth—and presents a credible theory of change linking the quantum-computing output to measurable development outcomes. Relevant beneficiaries, indicators, risks and possible unintended consequences are identified.
C6	8–10	Shows strong, plausible potential to advance relevant SDG but the causal pathway, measurement plan, indicators or beneficiary analysis need further development.
C6	4–7	Addresses a challenge related to relevant SDG, but the alignment and impact claims are broad, indirect or insufficiently substantiated.
C6	1–3	Mentions relevant SDG, or uses sustainable-development language, without demonstrating a credible contribution.
C6	0	Has no meaningful relevance to the relevant SDG.

SCORE AWARDED: _____ / 13

2.7 C7 – Feasibility, Scalability and Adoption (Maximum 10 points)

Can this be built, tested, improved and used?

C7	9–10	Provides a realistic implementation pathway, resource requirements, stakeholder roles, cost assumptions, risk assessment and staged scaling plan; quantum hardware or simulation requirements are realistic.
C7	7–8	Presents a credible pilot opportunity, although some operational, institutional, technical or financial assumptions remain unresolved.
C7	4–6	The solution is technically plausible but lacks a convincing deployment, maintenance, adoption or scaling pathway.
C7	1–3	Depends on unavailable data, inaccessible quantum hardware, unrealistic infrastructure, unsupported partnerships, excessive resources or untested assumptions.
C7	0	Cannot reasonably be implemented under foreseeable conditions.

SCORE AWARDED: _____ / 10

3.0 Scoring Adjustments

3.1 Evidence Confidence Multiplier

Confidence Level	Multiplier	When to Apply
HIGH	× 1.00	Claims are demonstrated, documented and reproducible.



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Confidence Level	Multiplier	When to Apply
MODERATE	× 0.95	Most claims are supported, but some important assumptions remain untested.
LOW	× 0.85	Important claims rely on projections, mock-ups, incomplete demonstrations or unclear evidence.

3.2 Quantum-Washing Penalty

Penalty	When to Apply
0 pts	The role of quantum computing is clearly justified and proportional to the claims made.
-3 pts	The quantum component is weakly justified or could easily be replaced by a classical technique.
-6 pts	The team makes unsupported claims of quantum speed-up, superiority or advantage.
-10 pts	Quantum terminology is primarily promotional, technically misleading or unrelated to the actual solution.

Final score formula: Final Score = (Raw Score × Evidence Confidence Multiplier) – Quantum-Washing Penalty

4.0 Complete Judging Sheet

Team Name		Project Title	
Judge Name		Date	
Track	Agriculture / Health / Finance / Climate / Other	Presentation No.	
Conflict of Interest Declared?	Yes / No	If yes, recusal required?	Yes / No

Criterion	Max	Score	Key Justification
C1 Problem Understanding and Local Relevance	15	—	
C2 Quantum Computing Relevance and Necessity	20	—	



Criterion	Max	Score	Key Justification
C3 Technical Correctness and Scientific Integrity	15	___	
C4 Evidence, Benchmarking and Validation	15	___	
C5 Innovation and Intellectual Contribution	12	___	
C6 Sustainable Development Impact (SDGs 2, 3 and 8)	13	___	
C7 Feasibility, Scalability and Adoption	10	___	
RAW SCORE	100	___	

4.1 Adjustment Selection

Adjustment	Selection	Result
Evidence confidence multiplier	×1.00 High / ×0.95 Moderate / ×0.85 Low	___
Quantum-washing penalty	0 / -3 / -6 / -10	___

FINAL SCORE: (Raw Score × Multiplier) – Penalty = _____ / 100